DTS103TC
Data Structure and Algorithms

Lab 2: Sorting and Searching

Dr. Qi Chen, Dr. Di Zhang, Dr. Md Maruf Hasan,
Dr. Xuechen Liu, Dr. Guangyu Ren and Dr. Bin Chen
School of AI and Advanced Computing

Learning outcomes

- Quicksort

- Sequential Search
- Binary Search

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Quicksort

Quicksort

- Divide-and-conquer algorithm
- Sorts "in place" (like insertion sort, but not like merge sort).

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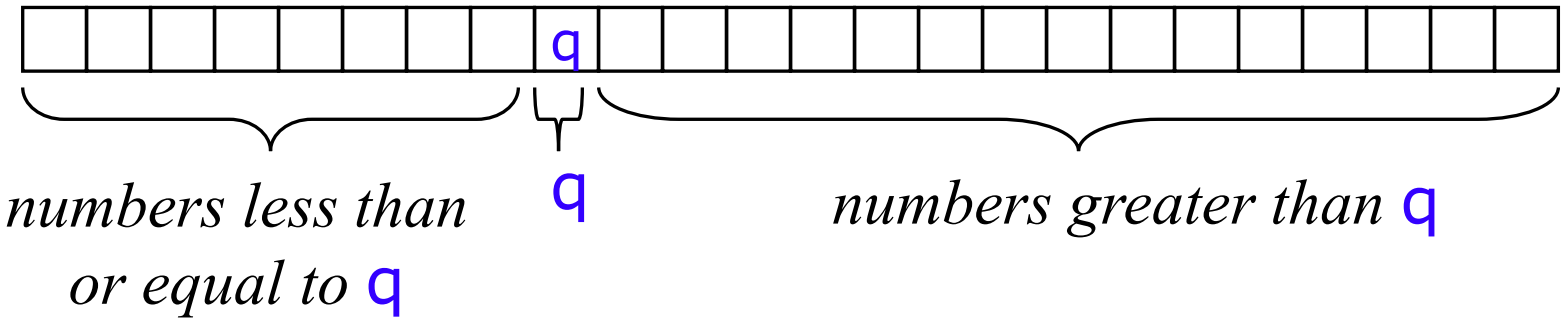
Quicksort

- Pick some number q from the array
- Move all numbers less than (or equal to) q to the beginning of the array
- Move all number greater than q to the end of the array
- Quicksort the numbers less than or equal to q
- Quicksort the numbers greater than or equal to q



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Quicksort pseudo code

```
Quciksort(A,p,r)
if p<r
        // partition an array such that:
        // all A[p,…,q-1] are less than or equal to A[q]
        // all A[q+1,…,r] are larger than A[q]
        q = Partition(A,p,r)
        Quicksort(A,p,q-1)
        Quicksort(A,q+1,r)
```

Initial call: Quicksort(A,1,n)

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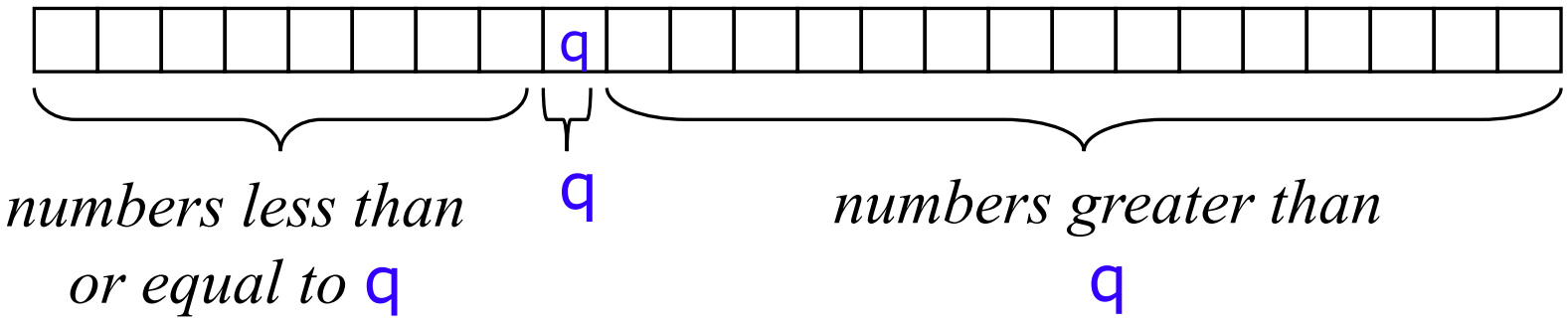
Partitioning

- A key step in the Quicksort algorithm is partitioning the array
  - We choose some (any) number q in the array to use as a pivot
  - We partition the array into three parts:



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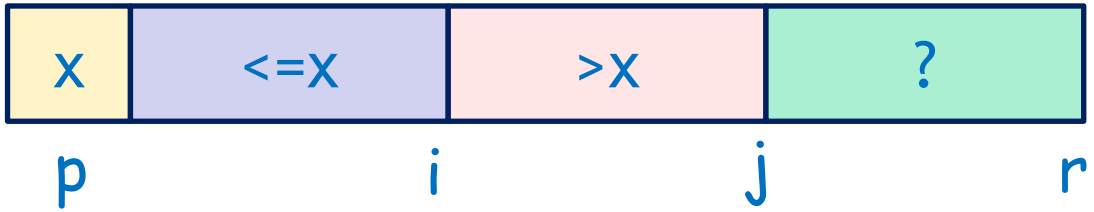
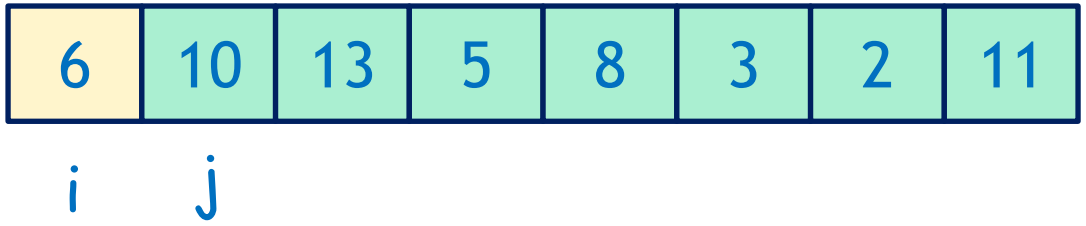
Partitioning




```
Partition(A,p,r)
x = A[p]
i = p
for j = p+1 to r
    if A[j] <= x
        i=i+1
        exchange A[i] with A[j]
exchange A[p] and A[i]
Return i
```



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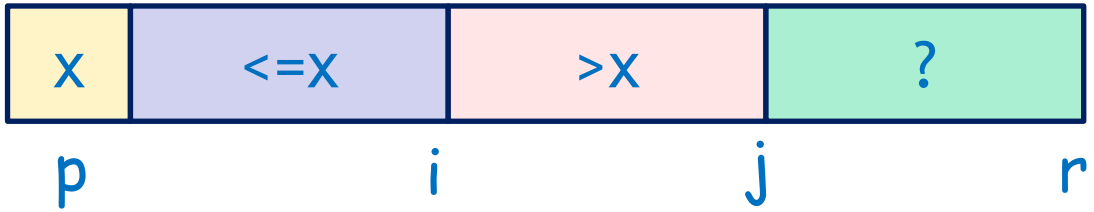
Partitioning

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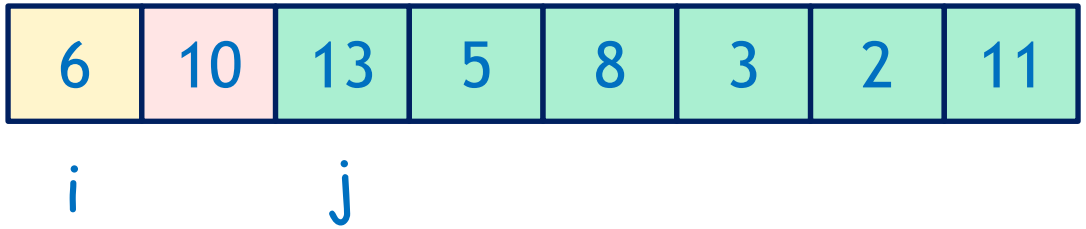

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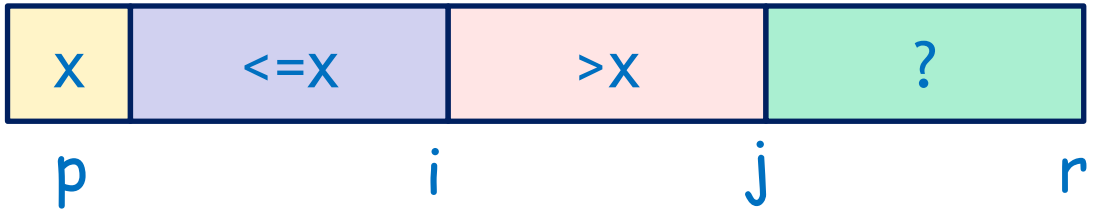


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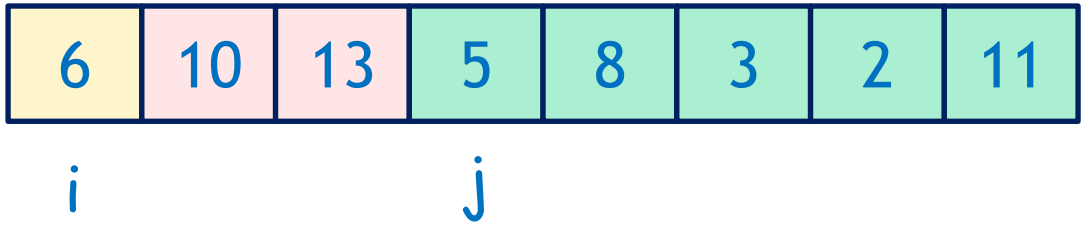

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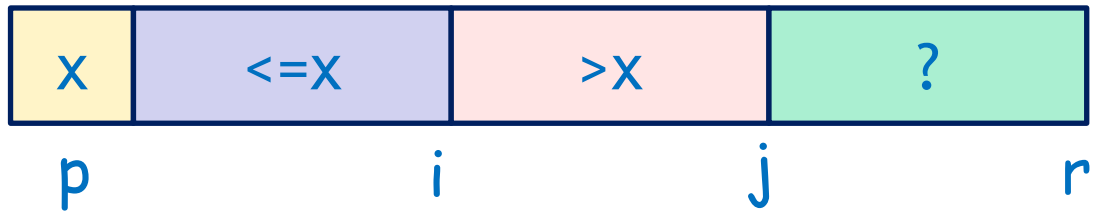

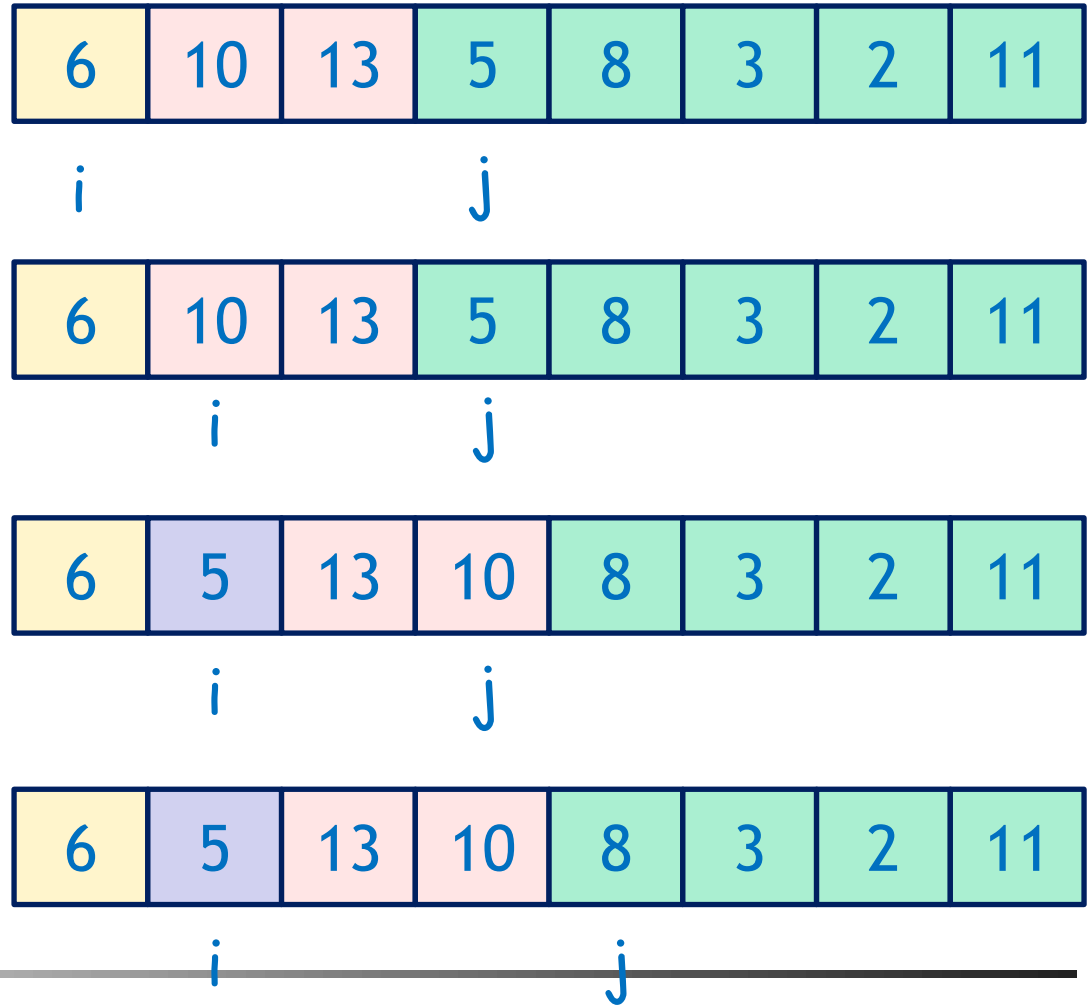

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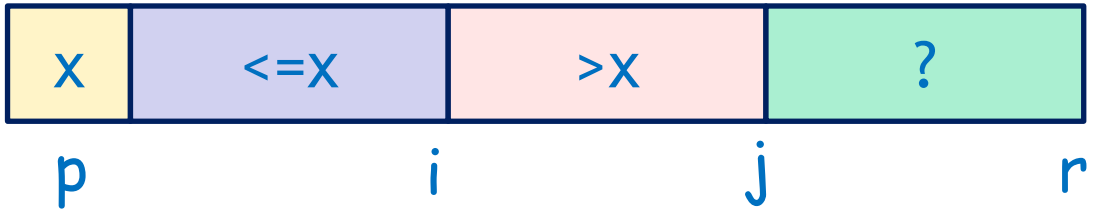
Partitioning

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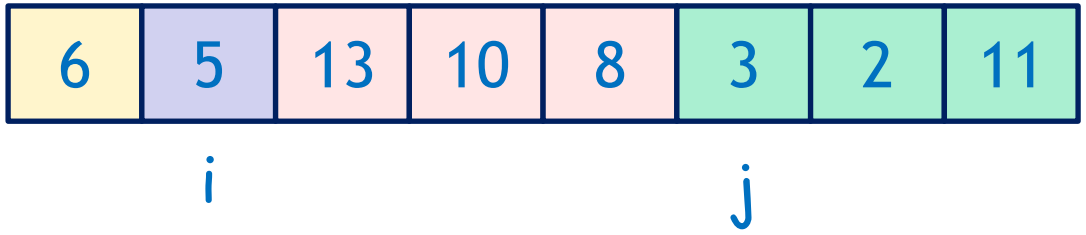

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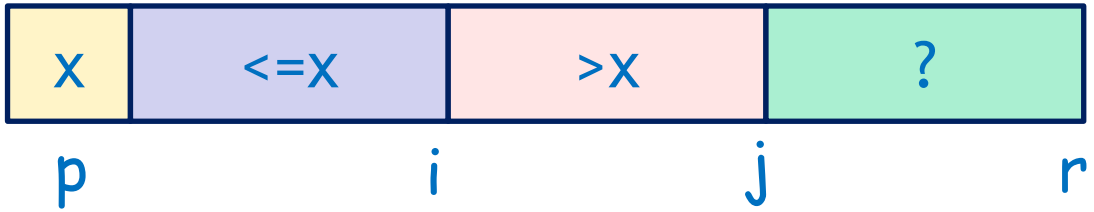

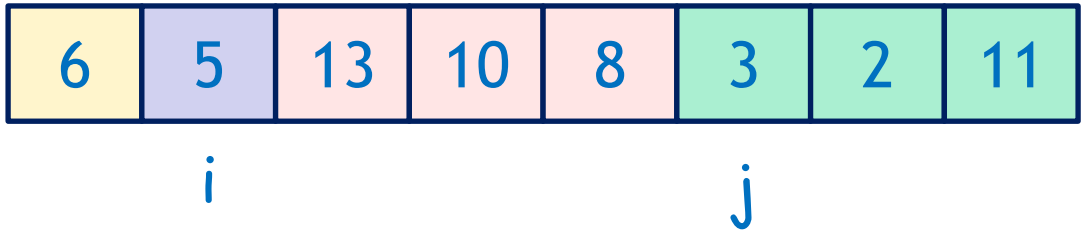

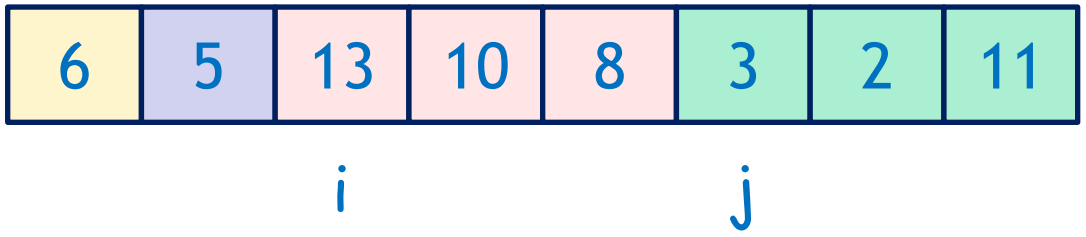

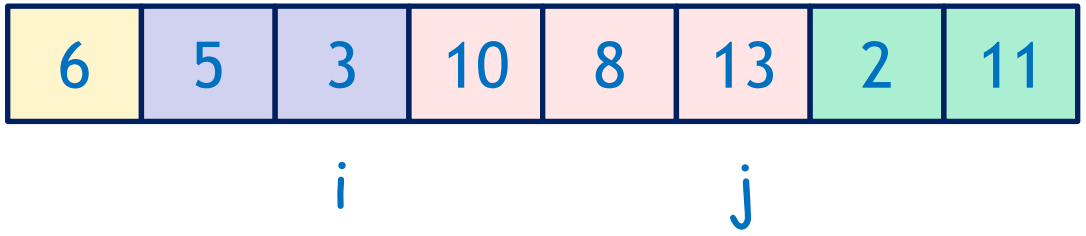

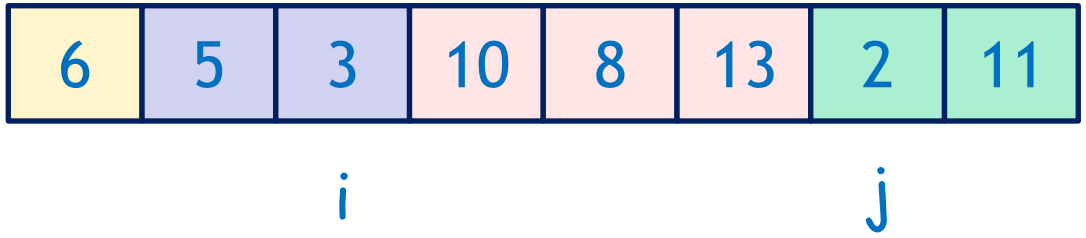

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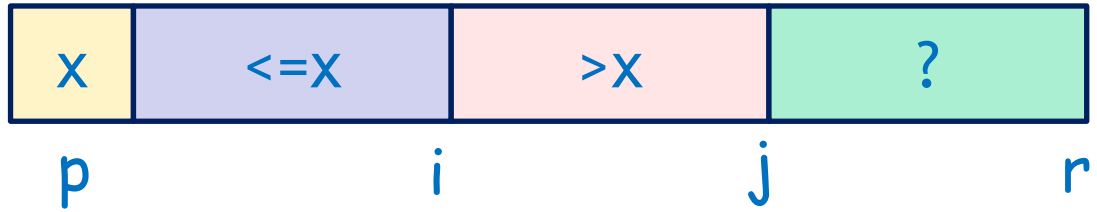

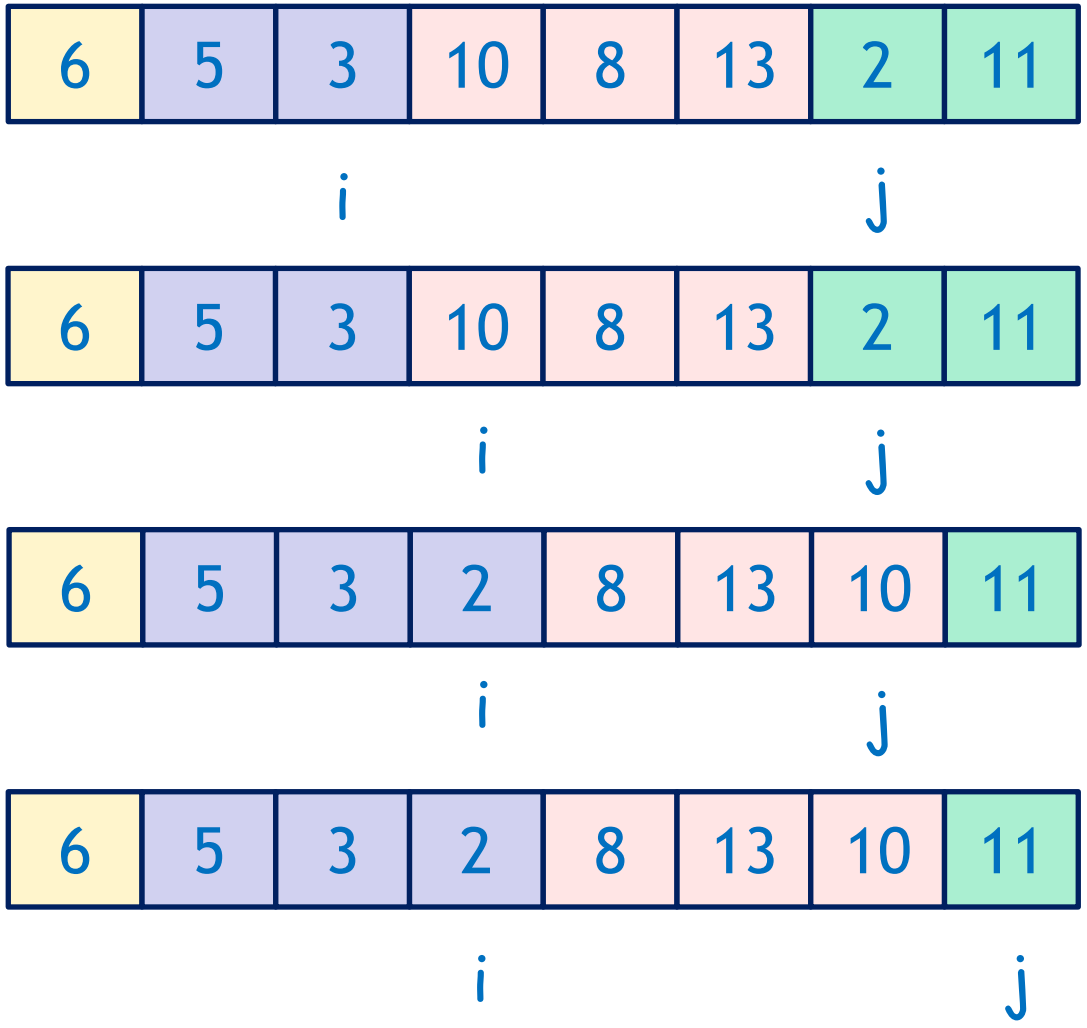

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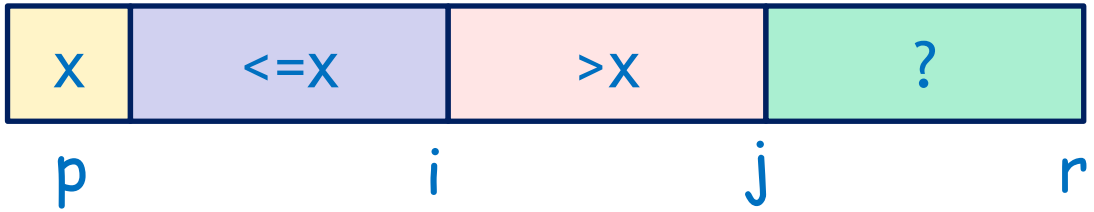

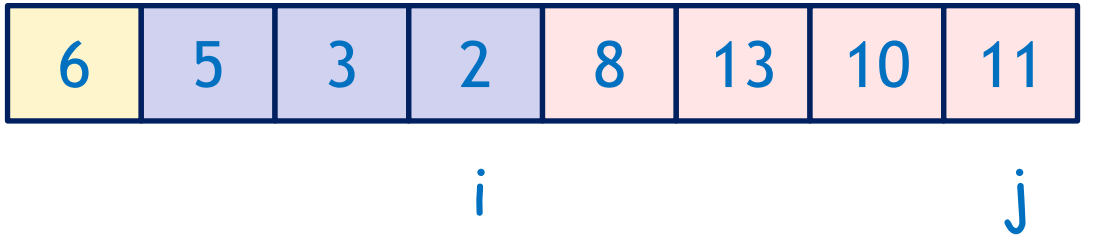

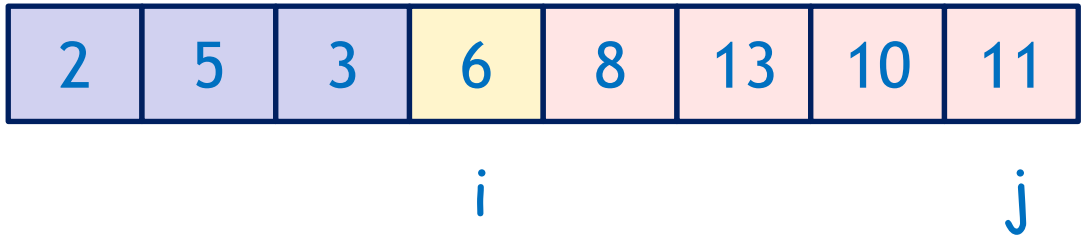

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Analysis of quicksort

- Best-case
  - Time complexity: $O(n\log n)$
  - Space complexity: $O(\log n)$

- Worst-case
  - Time complexity: $O(n^2)$
  - Space complexty: $O(n)$

- Average-case
  - Time complexity: $O(n\log n)$
  - Space complexity: $O(\log n)$

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Searching

Searching

- Input: n numbers $a_1, a_2, \ldots, a_n$ and a number X
- Output: determine if X is in the sequence or not

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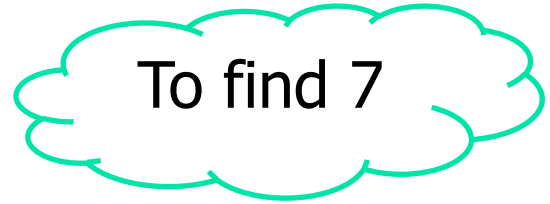
Sequential search

- 12 34 2 9 7 5
  7
- 12 34 2 9 7 5
  7
- 12 34 2 9 7 5
  7
- 12 34 2 9 7 5
  7
- 12 34 2 9 7 5
  7



found!

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Sequential search

- **12** 34 2 9 7 5
  10
- 12 **34** 2 9 7 5
  10
- 12 34 **2** 9 7 5
  10
- 12 34 2 **9** 7 5
  10
- 12 34 2 9 **7** 5
  10
- 12 34 2 9 7 **5**
  10 not found!

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Sequential search

```
i = 1
found = false
while (i<=n && found==false)
{
        if X == a[i] then
                found = true
        else
                i = i+1
}
```

Best case: X is 1st no.
$\Rightarrow$ 1 comparison $\Rightarrow O(1)$

Worst case: X is last OR X is not found $\Rightarrow$ n comparisons $\Rightarrow O(n)$

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How to improve Searching?

- Time complexity of Sequential searching is O(n).

- If a sorted array is given, can we improve the time complexity?

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Binary search

- Input: a sequence of n **sorted** numbers $a_1, a_2, \ldots, a_n$ in ascending order and a number X

- Idea of algorithm:
  - compare X with number in the middle
  - then focus on only the first half or the second half (depend on whether X is smaller or greater than the middle number)
  - reduce the amount of numbers to be searched by half

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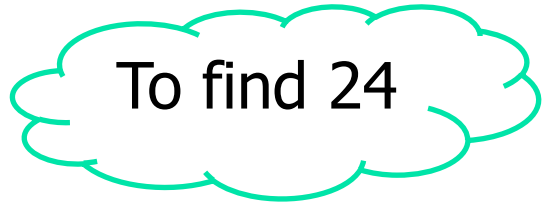
Binary Search



3 7 11 12 **15** 19 24 33 41 55
24

19 24 **33** 41 55
24

19 24
24

24
24

found!

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Binary Search

To find 30

3 7 11 12 **15** 19 24 33 41 55
30

19 24 **33** 41 55
30

19 24
30

24
30

not found!

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Binary Search – Pseudo Code

```
first = 1, last = n, found = false
while (first <= last && found == false)
{
        mid = ⌊(first+last)/2⌋
        if (X == a[mid])
                found = true
        else
                if (X < a[mid])
                        last = mid-1
                else
                        first = mid+1
}
if (found == true)
        report "Found"
else
        report "Not Found"
```

Best case: X is the number in the middle⇒ 1 comparison ⇒ $O(1)$

Worst case: at most (logn+1) comparisons ⇒ $O(\log n)$

Why?
Every comparison reduces the amount of numbers by at least half
E.g., 16 ⇒ 8 ⇒ 4 ⇒ 2 ⇒ 1

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Recursive Binary Search

```
RecurBinarySearch(A, first, last, X)
begin
   if (first > last) then
        return false
   mid = ⌊(first + last)/2⌋
   if (X == A[mid]) then
        return true
   if (X < A[mid]) then
        return RecurBinarySearch(A, first, mid-1, X)
   else
        return RecurBinarySearch(A, mid+1, last, X)
end
```

invoke by calling
RecurBinarySearch(A, 1, n, X)
return true if X is found,
false otherwise

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Exercises for Lab 2

- Implement Merge Sort (Check Lecture notes)
- Implement Quick Sort
- Implement Sequential Search
- Implement Binary Search
- Test the running time of different algorithms

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